



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr. Jayaram Boga
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Roll Number	2520030509
Branch	CSE
Course Outcome	CO-1

Case Study Topic: QuizClash Live Tournament Leaderboard using AVL Trees

Work Done Summary:

In this case study, we implemented an AVL Tree for the QuizClash live tournament leaderboard system.

The AVL Tree stores player scores in descending order and maintains subtree sizes for efficient rank queries.

We inserted player scores into the AVL Tree and observed balancing operations such as LL, RR, LR, and RL rotations.

We also implemented player rank retrieval and Top-K player queries in Java to support real-time leaderboard updates.

Implementation Details:

1. Inserted player scores into AVL Tree in arrival order.
2. Maintained balanced height after every insertion and deletion.
3. Performed AVL rotations whenever imbalance occurred.
4. Stored subtree size in every node for $O(\log n)$ rank calculation.
5. Implemented Top-K traversal for leaderboard display.
6. Simulated live score updates using delete + insert operations.
7. Compared AVL Tree performance with Linear Search and ArrayList.
8. Concluded AVL Trees improve live leaderboard efficiency because update and rank operations are $O(\log n)$. Result: Screenshots / Output

```
Initial Scores:
820 540 910 770 880 460 990 600 730 950 510

Updated Scores:
540 updated to 815
910 updated to 685

Top 5 Players:
990 950 880 820 815

Rank of 815: 4
Rank of 685: 7

Time Complexity:
AVL Insert/Delete:  $O(\log n)$ 
Rank Query:  $O(\log n)$ 
```

Compared to normal arrays and linear search methods, AVL Trees provide faster updates and rank retrieval for continuously changing multiplayer leaderboard systems.

GitHub link: <https://github.com/venkatalaxmisamhita/DSA-II-PROJ>

Student Signature	Faculty Signature